Syllabus Details

Syllabus ID:	13804
Syllabus Name:	**Working with Spring Framework_Làm việc với Spring Framework**
Course Name English:	**Working with Spring Framework**
Subject Code:	**HSF302**
Learning-Teaching Method:	Project based learning,Problem based learning,Active learning,In-class lecture
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	Study hour (150h) = 45h contact hours + 145-minute final exam + 102,6h self-study
Pre-Requisite:	PRJ301
Description:	Upon completion of this course students shoud: 1. Understand the followings: • Basic concepts of Java FX to build Desktop Application • Basics of Object-Relational Mapping (ORM) with JPA and how it simplifies database interactions. • Spring Framework, Spring Boot, which provides a comprehensive ecosystem for building enterprise-level applications. • Spring Framework's dependency injection, and web frameworks to build robust and scalable applications 2. Be able to: • Use JPA ORM tool to map Java objects to database tables. • Use Spring Framework's dependency injection, and web frameworks to build robust and scalable applications. • Develop Cross-platform Desktop applications and support for user experience (UI & UX) 3. Be able to work in team and present group's results
StudentTasks:	- Students must attend at least 80% of contact sessions in order to be accepted to the final examination. - Student is responsible to do all assigned exercises given by instructor in class or at home and submit on time - Use laptop in class only for learning purpose - Promptly access to the FU CMS at http://flm.fpt.edu.vn for up-to-date course information
Tools:	- Internet - IntelliJ IDEA - Platform : Java Development Kit 21 - Scene Builder 23
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	1402/QĐ-ĐHFPT dated 12/12/2025
IsApproved:	**True**
Note:	
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	12/12/2025

7 material(s)

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
1	JPA					N/A	True	False	True	https://docs.spring.io/spring-data/jpa/reference/jpa.html
2	Spring Framework Essentials		Spring Academy			N/A	False	False	True	https://spring.academy/courses/spring-framework-essentials

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
3	Spring Framework - Core Technologies		Spring Academy			N/A	True	False	True	https://docs.spring.io/spring-framework/reference/core.html
4	Spring Framework Specialization		Coursera			N/A	False	False	True	https://www.coursera.org/specializations/spring-framework
5	Java FX		Java FX			N/A	True	False	True	https://openjfx.io/openjfx-docs/
6	Spring 6 & Spring Boot 3 for Beginners		Udemy			N/A	False	False	True	https://www.udemy.com/course/learn-spring-boot/
7	Beginning Spring Data: Data Access and Persistence for Spring Framework 6 and Boot 3	Andres Sacco	Apress	29/12/2022	1st	1484287630	False	True	False	

6 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Describe the basics of Object-Relational Mapping (ORM) and how it simplifies database interactions.
2	CLO2	Describe the concepts of Spring Framework and Spring Boot
3	CLO3	Describe how Spring Data working with various data sources
4	CLO4	Undersatnd Spring MVC provides a comprehensive programming and configuration model for modern Java-based enterprise applications.
5	CLO5	Create Cross-Platform applications using JavaFX. JavaFX is platform-independent, which means that JavaFX applications (CRUD) can run on different operating systems such as Windows, macOS, and Linux
6	CLO6	Develop teamwork skills, communication skills (including presentation skills)

View mapping of CLOs to PLOs

Download All Student Material

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
1	Course Introduction	Offline	CLO1	IT	Slide	HSF302	- Read the slide, related reference documents - Do exercises - Self practice	
2	Working Environment	Offline	CLO1	TU	https://www.oracle.com/java/technologies/downloads/ https://www.eclipse.org/downloads/packages/ https://www.jetbrains.com/idea/download/ https://spring.io/tools https://gluonhq.com/products/scene-builder/ https://spring.io/tools		- Read the slide, related reference documents - Do exercises - Self practice	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
3	Project Introduction	Offline	CLO1	TU	Project List		- Choose Group - Choose Topic - Collect requirements about topic	
4	Chapter 01: JPA, JPA Mapping	Offline	CLO1	TU	https://docs.spring.io/spring-data/jpa/reference/jpa.html		- Read the slide, related reference documents - Do exercises - Self practice	
5	Chapter 01: JPA, JPA Mapping (contd.)	Offline	CLO1	TU	https://docs.spring.io/spring-data/jpa/reference/jpa.html		- Read the slide, related reference documents - Do exercises - Self practice	
6	Chapter 01: JPA, JPA Mapping (contd.)	Offline	CLO1	TU	https://docs.spring.io/spring-data/jpa/reference/jpa.html		- Read the slide, related reference documents - Do exercises - Self practice	
7	Chapter 01: JPA, JPA Mapping (contd.)	Offline	CLO1	TU	https://docs.spring.io/spring-data/jpa/reference/jpa.html		- Read the slide, related reference documents - Do exercises - Self practice	
8	Chapter 01: JPA, JPA Mapping (contd.)	Offline	CLO1	TU	https://docs.spring.io/spring-data/jpa/reference/jpa.html		- Read the slide, related reference documents - Do exercises - Self practice	
9	Chapter 01: JPA, JPA Mapping (contd.)	Offline	CLO1	TU	https://docs.spring.io/spring-data/jpa/reference/jpa.html		- Read the slide, related reference documents - Do exercises - Self practice	
10	Chapter 02: Introduction to Spring Framework	Offline	CLO2	TU	https://docs.spring.io/spring-framework/reference/core.html		- Read the slide, related reference document - Do exercises - Self practice	
11	Chapter 02: Introduction to Spring Framework (contd.)	Offline	CLO2	TU	https://docs.spring.io/spring-framework/reference/core.html		- Read the slide, related reference document - Do exercises - Self practice	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
12	Chapter 05: Introduction to Spring Framework (contd.)	Offline	CLO2	TU	https://docs.spring.io/spring-framework/reference/core.html		- Read the slide, related reference document - Do exercises - Self practice	
13	Chapter 03: Spring Boot Concepts	Offline	CLO2	ITU	https://docs.spring.io/spring-boot/docs/current/reference/html/ Beginning Spring Data: Data Access and Persistence for Spring Framework 6 and Boot 3: Chapter 02		- Read the slide, related reference documents - Do exercises - Self practice	
14	Chapter 03: Spring Boot Concepts (contd.)	Offline	CLO2	ITU	https://docs.spring.io/spring-boot/docs/current/reference/html/ Beginning Spring Data: Data Access and Persistence for Spring Framework 6 and Boot 3 : Chapter 02		- Read the slide, related reference documents - Do exercises - Self practice	
15	Chapter 03: Spring Boot Concepts (contd.)	Offline	CLO2	ITU	https://docs.spring.io/spring-boot/docs/current/reference/html/ Beginning Spring Data: Data Access and Persistence for Spring Framework 6 and Boot 3 : Chapter 02		- Read the slide, related reference documents - Do exercises - Self practice	
16	Chapter 03: Spring Boot Concepts (contd.)	Offline	CLO2	ITU	https://docs.spring.io/spring-boot/docs/current/reference/html/ Beginning Spring Data: Data Access and Persistence for Spring Framework 6 and Boot 3 : Chapter 03		- Read the slide, related reference documents - Do exercises - Self practice	
17	Chapter 03: Spring Boot Concepts (contd.)	Offline	CLO2	ITU	https://docs.spring.io/spring-boot/docs/current/reference/html/ Beginning Spring Data: Data Access and Persistence for Spring Framework 6 and Boot 3 : Chapter 04		- Read the slide, related reference documents - Do exercises - Self practice	
18	Chapter 03: Spring Boot Concepts (contd.)	Offline	CLO2	ITU	https://docs.spring.io/spring-boot/docs/current/reference/html/ Beginning Spring Data: Data Access and Persistence for Spring Framework 6 and Boot 3 : Chapter 05		- Read the slide, related reference documents - Do exercises - Self practice	
19	Chapter 04. Spring Data JPA	Offline	CLO3	IT	https://docs.spring.io/spring-data/jpa/reference/ Beginning Spring Data: Data Access and Persistence for Spring Framework 6 and Boot 3 : Chapter 04		- Read the slide, related reference documents - Do exercises - Self practice	
20	Chapter 04. Spring Data JPA (contd.)	Offline	CLO3	TU	https://docs.spring.io/spring-data/jpa/reference/ Beginning Spring Data: Data Access and Persistence for Spring Framework 6 and Boot 3 : Chapter 04		- Read the slide, related reference documents - Do exercises - Self practice	
21	Chapter 04. Spring Data JPA (contd.)	Offline	CLO3	TU	https://docs.spring.io/spring-data/jpa/reference/ Beginning Spring Data: Data Access and Persistence for Spring Framework 6 and Boot 3 : Chapter 04		- Read the slide, related reference documents - Do exercises - Self practice	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
22	Chapter 04. Spring Data JPA (contd.)	Offline	CLO3	TU	https://docs.spring.io/spring-data/jpa/reference/ Beginning Spring Data: Data Access and Persistence for Spring Framework 6 and Boot 3 : Chapter 04		- Read the slide, related reference documents - Do exercises - Self practice	
23	Chapter 04. Spring Data JPA (contd.)	Offline	CLO3	TU	https://docs.spring.io/spring-data/jpa/reference/ Beginning Spring Data: Data Access and Persistence for Spring Framework 6 and Boot 3 : Chapter 04		- Read the slide, related reference documents - Do exercises - Self practice	
24	Chapter 04. Spring Data JPA (contd.)	Offline	CLO3	TU	https://docs.spring.io/spring-data/jpa/reference/ Beginning Spring Data: Data Access and Persistence for Spring Framework 6 and Boot 3 : Chapter 04		- Read the slide, related reference documents - Do exercises - Self practice	
25	Guiding Assignment 01	Offline	CLO2	TU	Assignment 01		- Read the requirements of Assignment 01 - Self practice	
26	Progress Test 01	Offline	CLO2	TU	Quiz LMS		- Take Quiz 01	
27	Project Assistant - 01	Offline	CLO2	TU	Group Project List		- Discuss project problems	
28	Chapter 05: Spring MVC	Offline	CLO3	TU	https://docs.spring.io/spring-framework/reference/core.html		- Read the slide, related reference document - Do exercises - Self practice	
29	Chapter 05: Spring MVC (contd.)	Offline	CLO3	TU	https://docs.spring.io/spring-framework/reference/core.html		- Read the slide, related reference document - Do exercises - Self practice	
30	Chapter 05: Spring MVC (contd.)	Offline	CLO3	TU	https://docs.spring.io/spring-framework/reference/core.html		- Read the slide, related reference document - Do exercises - Self practice	
31	Guiding Assignment 02	Offline	CLO3	TU	Assignment 02		- Read the requirements of Assignment 02 - Self practice	
32	Progress Test 02	Offline	CLO3	TU	Quiz LMS		- Take Quiz 02	
33	Project Assistant - 02	Offline	CLO3	TU	Group Project List		- Discuss project problems	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
34	Chapter 05: Spring MVC (contd.)	Offline	CLO4	TU	https://docs.spring.io/spring-framework/reference/core.html		- Read the slide, related reference document - Do exercises - Self practice	
35	Chapter 05: Spring MVC (contd.)	Offline	CLO4	TU	https://docs.spring.io/spring-framework/reference/core.html		- Read the slide, related reference document - Do exercises - Self practice	
36	Chapter 05: Spring MVC (contd.)	Offline	CLO4	TU	https://docs.spring.io/spring-framework/reference/core.html		- Read the slide, related reference document - Do exercises - Self practice	
37	Chapter 06: Spring Web Development with Thymeleaf	Offline	CLO4	TU	https://docs.spring.io/spring-framework/reference/core.html		- Read the slide, related reference document - Do exercises - Self practice	
38	Chapter 06: Spring Web Development with Thymeleaf (contd.)	Offline	CLO4	TU	https://docs.spring.io/spring-framework/reference/core.html		- Read the slide, related reference document - Do exercises - Self practice	
39	Chapter 06: Spring Web Development with Thymeleaf (contd.)	Offline	CLO4	TU	https://docs.spring.io/spring-framework/reference/core.html		- Read the slide, related reference document - Do exercises - Self practice	
40	Chapter 06: Spring Web Development with Thymeleaf (contd.)	Offline	CLO4	TU	https://docs.spring.io/spring-framework/reference/core.html		- Read the slide, related reference document - Do exercises - Self practice	
41	Chapter 06: Spring Web Development with Thymeleaf (contd.)	Offline	CLO4	TU	https://docs.spring.io/spring-framework/reference/core.html		- Read the slide, related reference document - Do exercises - Self practice	
42	Chapter 06: Spring Web Development with Thymeleaf (contd.)	Offline	CLO4	TU	https://docs.spring.io/spring-framework/reference/core.html		- Read the slide, related reference document - Do exercises - Self practice	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
43	Chapter 07: Build a simple application with Data Access using Spring & Spring JDBC, Spring ORM	Offline	CLO4	TU	https://docs.spring.io/spring-framework/reference/core.html		- Read the slide, related reference document - Do exercises - Self practice	
44	Chapter 07: Build a simple application with Data Access using Spring & Spring JDBC, Spring ORM (contd.)	Offline	CLO4	TU	https://docs.spring.io/spring-framework/reference/core.html		- Read the slide, related reference document - Do exercises - Self practice	
45	Chapter 07: Build a simple application with Data Access using Spring & Spring JDBC, Spring ORM (contd.)	Offline	CLO4	TU	https://docs.spring.io/spring-framework/reference/core.html		- Read the slide, related reference document - Do exercises - Self practice	
46	Chapter 07: Build a simple application with Data Access using Spring & Spring JDBC, Spring ORM (contd.)	Offline	CLO4	TU	https://docs.spring.io/spring-framework/reference/core.html		- Read the slide, related reference document - Do exercises - Self practice	
47	Chapter 07: Build a simple application with Data Access using Spring & Spring JDBC, Spring ORM (contd.)	Offline	CLO4	TU	https://docs.spring.io/spring-framework/reference/core.html		- Read the slide, related reference document - Do exercises - Self practice	
48	Chapter 07: Build a simple application with Data Access using Spring & Spring JDBC, Spring ORM (contd.)	Offline	CLO4	TU	https://docs.spring.io/spring-framework/reference/core.html		- Read the slide, related reference document - Do exercises - Self practice	
49	Chapter 08: Introduction to JavaFX	Offline	CLO5	TU	https://openjfx.io/openjfx-docs/		- Read the slide, related reference documents - Do exercises - Self practice	
50	Chapter 08: Introduction to JavaFX (contd.)	Offline	CLO5	TU	https://openjfx.io/openjfx-docs/		- Read the slide, related reference documents - Do exercises - Self practice	
51	Chapter 08: Introduction to JavaFX (contd.)	Offline	CLO5	TU	https://openjfx.io/openjfx-docs/		- Read the slide, related reference documents - Do exercises - Self practice	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
52	Chapter 08: Introduction to JavaFX (contd.)	Offline	CLO5	TU	https://openjfx.io/openjfx-docs/		- Read the slide, related reference documents - Do exercises - Self practice	
53	Chapter 08: Introduction to JavaFX (contd.)	Offline	CLO5	TU	https://openjfx.io/openjfx-docs/		- Read the slide, related reference documents - Do exercises - Self practice	
54	Chapter 08: Introduction to JavaFX (contd.)	Offline	CLO5	TU	https://openjfx.io/openjfx-docs/		- Read the slide, related reference documents - Do exercises - Self practice	
55	Chapter 09: Build application with JavaFX	Offline	CLO5	TU	https://openjfx.io/openjfx-docs/		- Read the slide, related reference documents - Do exercises - Self practice	
56	Chapter 09: Build application with JavaFX (contd.)	Offline	CLO5	TU	https://openjfx.io/openjfx-docs/		- Read the slide, related reference documents - Do exercises - Self practice	
57	Chapter 09: Build application with JavaFX (contd.)	Offline	CLO5	TU	https://openjfx.io/openjfx-docs/		- Read the slide, related reference documents - Do exercises - Self practice	
58	Project Evaluation	Offline	CLO1, CLO2, CLO3, CLO4	TU	https://docs.spring.io/spring-framework/reference/core.html		- Read the slide, related reference document - Do exercises - Self practice	
59	Project Evaluation	Offline	CLO1, CLO2, CLO3, CLO4	TU	https://docs.spring.io/spring-framework/reference/core.html		- Read the slide, related reference document - Do exercises - Self practice	
60	Project Evaluation	Offline	CLO1, CLO2, CLO3, CLO4	TU	https://docs.spring.io/spring-framework/reference/core.html		- Read the slide, related reference document - Do exercises - Self practice	

0 Constructive question(s)

6 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
1	Assignment	on-going	2	10.0%	>0	12 slots	CLO2, CLO3, CLO4			Subjects in syllabus		
2	Group Project	on-going	1	25.0%	>0	20 slots	CLO2, CLO3, CLO4, CLO5			Subjects in syllabus		
3	Progress test	on-going	2	10.0%	>0	30'	CLO1, CLO2, CLO3, CLO4	Multiple choices Marked by Computer or a suitable format	20/each			
4	Final exam	Final exam	2	55.0%	4	145 minutes	CLO1, CLO2, CLO3, CLO4, CLO5			All subjects in syllabus	By exam board Manual evaluating	Each part of the final exam must be >=4/10 & Final Result >=5/10.
5	Practical Exam	Final exam	1	25.0%	4	85'	CLO2, CLO3, CLO4			Subjects in syllabus	By exam board Manual evaluating	
6	Theory Exam	Final exam	1	30.0%	4	60'	CLO1, CLO2, CLO3, CLO4, CLO5		50	All subjects in syllabus	By exam board	